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In-Game Behavior and Performance: A Comparative Network Analysis of Esports Performance Among Professional Counter-Strike Players 2012–2025

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ABSTRACT

In esports, competitive performance is directly reflected in players' in-game behavior. However, the interrelationships among in-game behavior and performance still remain inadequately understood. First-person shooter (FPS) games provide an important context for studying the organizational structure and interactions of in-game behavior in esports. This study employs network analysis to explore 39 in-game behavioral indicators among professional Counter-Strike players across 1,468 players from CS:GO (2012–2023) and 880 players from CS:2 (2023–2025). The research objective is to construct and explore the network structure, invariance, and community characteristics of in-game behavioral networks. Results showed that stronger associations primarily occurred within the same in-game behavioral dimensions, while some cross-dimensional connections were also observed. Certain core in-game behaviors maintained a high level of node strength and expected influence across both periods, particularly metrics related to the Entrying, Trading, and Firepower dimensions. The rankings of the relative importance of bridging strength and predictability for in-game behaviors varied between CS:GO and CS2. Network Comparison Test (NCT) results revealed significant differences between the two periods in both global network strength and network structure. Stability tests confirmed the reliability of the network analysis. Additionally, the behavioral dimensions identified through community detection differ from the original dimensions. This study provides empirical evidence for understanding the interrelationships among in-game behaviors in esports. It offers a potential analytical perspective for enhancing esports athletes' performance, training feedback, and tactical management, while also advancing interdisciplinary development in human-computer interaction and esports.

KEYWORDS

Esports performance; in-game behavior; network analysis; counter-strike; esports ecosystems

1. Introduction

The digital environment of esports allows us to observe players' gaming behaviors and performance across different periods within the esports ecosystem (Kriglstein et al., 2021). In this context, the various in-game behaviors exhibited in esports rarely exist in isolation (Nagorsky & Wiemeyer, 2020). In this study, in-game behavior refers to players' measurable performance-related actions and outcomes within the game system, as captured by official match statistics reflecting combat execution, opening engagement, trading, utility use, survival, and clutch-related play (Jing et al., 2025). These behaviors typically manifest within a game ecosystem shaped by specific match scenarios, tactical combinations, and team role interactions (Thaicharoen et al., 2023). When the game's background conditions change such as balance adjustments or version updates, players' performance and specific in-game behavioral statistics may fluctuate (Zhang & Keegan, 2022). At the same time, the associations of in-game behaviors may undergo organizational restructuring, in which the relationships between different tactical

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capabilities and behavioral chains are reconfigured (Thaicharoen et al., 2023; Zhang & Keegan, 2022). Therefore, esports serves not only as a platform for competitive performance, but also as an exceptional human-computer interaction environment for understanding how players act, make decisions and adjust their in-game behavior within digital systems (Kriglstein et al., 2021).

Within the FPS genre and the broader esports landscape, Counter-Strike (CS) stands as one of the most established and influential esports games, offering a highly structured and refined digital competitive environment (Sadowska et al., 2023). Furthermore, CS has evolved into a professional ecosystem similar to traditional sports, featuring a formal coaching system, institutionalized training and rigorous data analysis. This game characterized by its round-based objective missions, economic management system, and specialization of roles (Jing et al., 2025). During each round, teams must vie for spatial dominance and accomplish offensive or defensive objectives, with their in-game behavior typically exhibiting distinct characteristics across different phases (Young, 2021). For example, the opening phase focuses on probing and intelligence gathering, the mid-game emphasizes tactical decision-making, firepower output and utility exchanges, while the late phase centers on endgame scenarios (Jing et al., 2025). Unlike traditional sports, CS's economic system integrates different weapons and utilities (e.g., smoke grenades, flashbangs, etc.) into gameplay, making them critical elements for team coordination and breaking through stalemates (Xia et al., 2025). Additionally, the performance of professional teams heavily relies on role specialization and teamwork, such as offensive players, leaders, and snipers (Jing et al., 2025). These roles carry specific responsibilities in different match scenarios, enabling in-game behavior to reflect not only individual skill but also their role orientation and task allocation within teamwork (HLTV, 2023). Previous studies have utilized data from the HLTV platform for esports research (Irwin et al., 2025; Rejthar & Kotrba, 2023). Therefore, this competitive ecosystem encompassing the complex and interrelated gaming actions within the digital environment provides an ideal setting for exploring the evolution of esports performance and the interaction of in-game behaviors.

Compared to traditional sports, research on esports performance remains quite limited (Białecki et al., 2024; Kriglstein et al., 2021). Existing research on esports players primarily focuses on factors related to body condition, physiological responses, and mental health (McNulty et al., 2023; Watanabe et al., 2021). However, these external physiological states or manifestations are not behavioral expressions within the game environment itself. In esports, competitive performance is directly reflected through specific in-game behavior and performance metrics, and these in-game actions more directly reveal how players interact with the game, adapt tactics, and collaborate with their team (Nagorsky & Wiemeyer, 2020). Therefore, in addition to the physiological perspective, it remains necessary to study the organizational structure and interrelationships of in-game behavioral metrics, particularly across different game phases and competitive ecosystems. There are also some gaps in the field of esports research related to in-game behavior and performance. First, previous studies focus on predicting match results (Xenopoulos et al., 2020; Xia et al., 2025), it failed to reveal the conditional dependencies among key indicators. Specifically, it overlooks how different in-game behavior indicators collectively form a behavioral network or chain. Secondly, game performance may be influenced by the combined effects of actual match scenarios, economic management, and team interactions (Nagorsky & Wiemeyer, 2020; Thaicharoen et al., 2023). Statistical relationships between data might form in-game behavior modules tailored to specific game scenarios. Moreover, in-game behavior may also be affected by changes introduced in version updates (Thaicharoen et al., 2023). This limitation is particularly pronounced in FPS esports, major changes to the game environment and tactical systems would reshape the individual performance and in-game behavior relationships. It is because performance in esports does not arise outside the digital environment, as it is driven by repetitive in-game actions embedded within tactical contexts and teamwork mechanics (Nagorsky & Wiemeyer, 2020). Additionally, compared to amateur players, elite players' performance is more heavily influenced by factors such as teamwork and highly complex micro-maneuvers. The in-game behavior and performance metrics of elite players hold greater practical significance for the development and training of future esports athletes, as well as for the formulation of high-level tactics (Jing et al., 2025). Therefore, we must transcend isolated behavioral analysis and delve deeper into the relational structures and systemic dynamics underlying

these interdependent in-game behaviors. At this stage, several issues remain inadequately clarified, including player performance, behavioral interactions, and the evolution of related tactical trends.

The research paradigm of network analysis offers us a new perspective for exploring in-game behavior. Network analysis is an analytical paradigm based on graph theory, relying on computational and statistical modeling (Borgatti et al., 2009; Borsboom et al., 2021; Borsboom & Cramer, 2013). This paradigm abstracts research subjects into a system of nodes and edges, thereby enabling the exploration of relational structures among individuals and the systemic characteristics that emerge from these structures (Borsboom, 2017). This perspective provides an analytical framework for understanding the dependencies and organizational structures of in-game behaviors, and has been applied in psychological symptom network and football network (Jing et al., 2026; Pan et al., 2025; Yi et al., 2026). Through network construction, we treat individual in-game behavior indicators as nodes, with their conditional associations serving as edges. This framework identifies network structures, core network metrics, node predictability and community detection (Borsboom, 2017; Haslbeck & Fried, 2017; Robinaugh et al., 2016; Yang et al., 2016), thereby revealing structural associations and dimensional divisions within the network of in-game behavior indicators. Additionally, Network Comparison Testing (NCT) facilitates the formal evaluation of connectivity and topological differences across various networks (van Borkulo et al., 2023). Consequently, by examining the associations and structural reorganization of in-game behaviors from the CS:GO to CS2 period, we aim to provide empirical insights into the topological evolution of player performance within the esports ecosystem.

In summary, this study employs network analysis to explore the structural organization of professional esports players' in-game behaviors across different periods. Its objective is to describe and compare the in-game behavior networks of players during the CS:GO (2012–2023) and the CS2 (2023–2025). Our research questions revolve around the following: (1) Are there significant differences in the overall network structure and strength between these two periods? (2) Which in-game behaviors occupy central topological positions within the network? (3) Do the core nodes and structure change across these two periods? (4) Do community detection results align with dimensions of original in-game behavior? By exploring these issues, this study helps us understand the evolution and restructuring of in-game behavioral patterns underlying professional esports performance within the digital esports' ecosystem. Furthermore, our findings will provide an empirical foundation and analytical pathways for subsequent esports performance evaluation, scientific training, in-game behavioral chain diagnosis and game tactical analysis.

2. Method

2.1. Research design

This study serves as a follow-up to a previous research examining the esports performance of professional Counter-Strike players from 2012 to 2025 (Jing et al., 2025). The dataset encompasses career statistics for 1,468 professional players during the CS:GO period (2012–2023) and 880 professional players during the CS2 period (2023–2025). The 39 official HLTV in-game behavior indicators covering seven dimensions (Firepower, Entrying, Trading, Opening, Clutching, Sniping, and Utility) were included, and descriptive information can be found in the supplementary materials. Previous research focused on interpretable machine learning frameworks for outcome-oriented prediction and indicator importance interpretation. This study focuses on exploring the associative structure among in-game behavior indicators and the evolution of topological features within behavior indicator networks through a network analysis framework. These two analytical frameworks address different research questions and directions. Therefore, network analysis provides complementary evidence for understanding how professional esports players' in-game behaviors are organized across different periods and their potential structural reconfigurations. The data used in this study is derived from official player career statistics provided by HLTV, these publicly available statistics were calculated separately for each period (CSGO/CS2). To eliminate potential bias from insufficient match data or excessive volatility, the final dataset includes only professional players who have participated in at least 100 official maps throughout their careers. This ensures that the in-game behavioral metrics are derived from meaningful professional

match data. The specific data collection process can be detailed in prior research (Jing et al., 2025). The data for this study were sourced from the publicly available HLTV database and did not involve human experiments.

2.2. Measures

The in-game behavior indicators in this study are derived from HLTV's official advanced statistics system, specifically encompassing seven dimensions. Firepower: Kills per round (KPR), Kills per round win (KPRW), Damage per round (DPR), Damage per round win (DPRW), Rounds with a kill (RWK), Rounds with a multi-kill (RWMK), Pistol round rating (PRR); Entry: Saved by teammate per round (SBTPR), Traded deaths per round (TDPR), Traded deaths percentage (TDP), Opening deaths traded percentage (ODTP), Assists per round (ASPR), Support rounds (SR); Trading: Saved teammate per round (STPR), Trade kills per round (TKPR), Trade kills percentage (TKP), Assisted kills percentage (AKP), Damage per kill (DPK); Clutching: Clutch points per round (CPPR), Last alive percentage (LAP), 1on1 win percentage (WP1V1), Time alive per round (TAPR), Saves per round loss (SPRL), Sniper kills per round (SKPR); Opening: Opening kills per round (OKPR), Opening deaths per round (ODPR), Opening attempts (OA), Opening success (OS), Win% after opening kill (WAOK), Attacks per round (ATPR); Sniping: Sniper kills percentage (SKP), Rounds with sniper kills percentage (RWSKP), Sniper multi-kill rounds (SMKR), Sniper opening kills per round (SOKPR); Utility: Utility damage per round (UDPR), Utility kills per 100 rounds (UKPR100), Flashes thrown per round (FTPR), Flash assists per round (FAPR), Time opponent flashed per round (TOFPR). The official definition of these in-game behavior indicators will be provided in supplementary materials.

2.3. Data analysis

2.3.1. Network analysis

The network analysis process as shown in Figure 1. All network statistic and visualizations were performed in R (R Core Team, 2013). When constructing the network of in-game behavior indicators, each indicator was treated as a node, and statistical associations between indicators were treated as edges. This study estimated the conditional association structure between indicators based on the partial

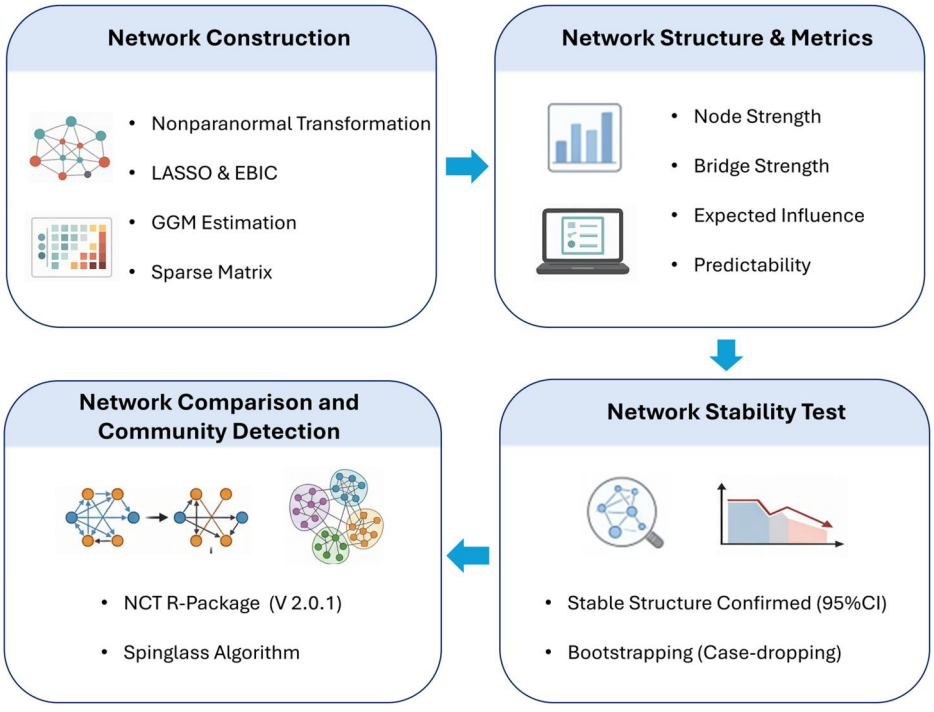


Figure 1. Network analysis and design process.

correlation matrix to characterize the independent association between any two indicators after controlling for other indicators. We applied a nonparanormal transformation to reduce the influence of skewness and extreme values on association estimates (Bai et al., 2021). We then used the graphical least absolute shrinkage and selection operator (LASSO) to regularize and sparsify the partial correlation matrix, yielding a more parsimonious and interpretable network (Tibshirani, 1996). Model selection and tuning were performed using the extended Bayesian information criterion (EBIC) (Haslbeck & Waldorp, 2018). Network estimation, stability analyses, and visualization were implemented primarily with the R packages bootnet and qgraph (Epskamp et al., 2018). Edge width reflects the association weighted. Green edges indicate positive associations, red edges indicate negative associations.

Network metrics include node strength, bridge strength, predictability, and expected influence. Node strength summarizes the overall strength of connections between nodes and serves as a measure of centrality (Epskamp et al., 2012). Bridging strength represents the degree to which a node connects different communities. Nodes with high bridging strength serve as critical hubs linking distinct modules (Jones et al., 2021). Predictability measures the proportion of variation in a node that can be explained by its neighboring node (Haslbeck & Waldorp, 2018). Expected influence measures the weighted association between a node and its neighboring nodes, accounting for the positive and negative directions of edges to characterize the node's net influence within the network (Robinaugh et al., 2016).

2.3.2. Network stability

All network stability tests are performed with the bootnet package (Version 1.4.3) (Epskamp, 2018). This study focuses on the stability of the most representative network centrality metric, specifically the performance of node strength and bridging strength (Epskamp et al., 2018). Network stability is determined using the correlation stability coefficient (CS-C) to identify the maximum proportion of cases that can be removed while maintaining a correlation higher than 0.7 with the original centrality metrics (Epskamp et al., 2018). CS-C value exceeding 0.25 meets the minimum requirement, while a CS-C value above 0.5 indicates higher stability (Bai et al., 2021; Epskamp et al., 2018). Additionally, we employed nonparametric bootstrapping to derive 95% confidence intervals (CIs) for edge weights, with narrower intervals indicating greater parametric certainty (Epskamp et al., 2018).

2.3.3. Comparison of network

We compared the CS:GO and CS2 networks using the Network Comparison Test (NCT) implemented in the R package NetworkComparisonTest (v2.0.1) (van Borkulo et al., 2015), which applies a permutation-based procedure to evaluate between-network differences. Following standard NCT outputs, we tested: (a) global strength invariance, indicating whether overall connectivity differs between networks; (b) network structure invariance, indicating whether the overall configuration of edges differs; and (c) edge invariance, identifying specific edges that differ across networks.

2.3.4. Network communities

We identified communities using the spinglass algorithm, which detects densely connected clusters by optimizing a modularity function and can capture hierarchical community structure (Yang et al., 2016). Spinglass has been used in previous network studies and has shown acceptable reliability in community detection (Moshier et al., 2018; Yuan et al., 2022).

3. Result

3.1. Network structure

The descriptive summary of in-game behavioral metrics can be found in the supplementary materials. The network of in-game behavior network during the CSGO and CS2 period is shown in Figure 2 and Figure 4. We present the top 20 strongly associated edges within the networks during CSGO and CS2 period.

Top ten strongly associated edges of the in-game behavior network during the CSGO period followed by TKPR—TKP (Trading-Trading), TDPR—TDP (Entrying—Entrying), FTPR—FAPR (Utility—Utility), UDPR—UKPR100 (Utility—Utility), OKPR—OA (Opening—Opening), FAPR—TOFPR

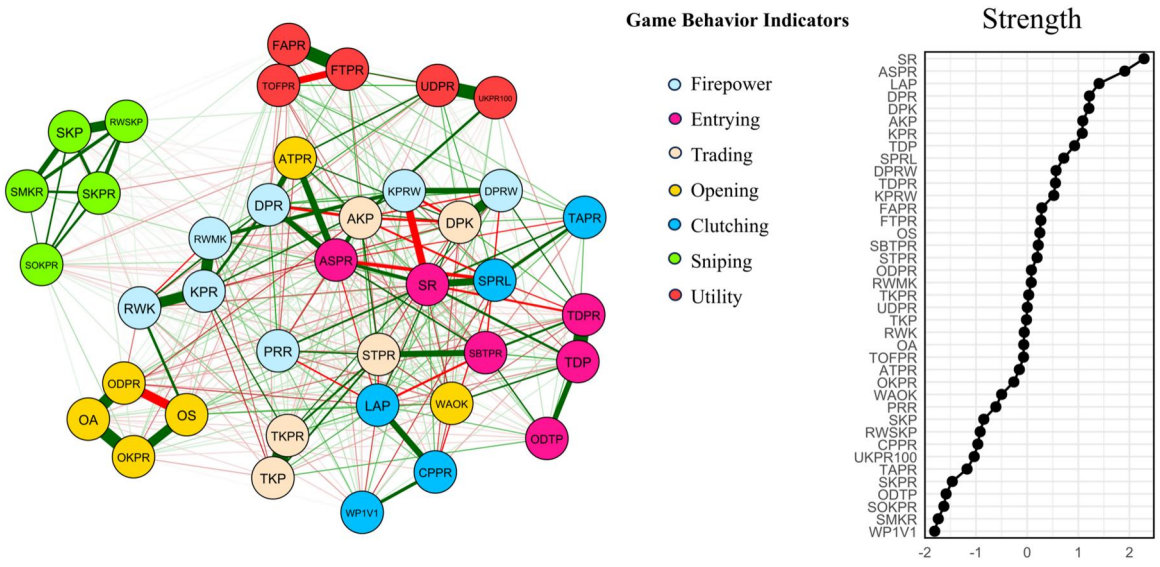


Figure 2. In-game behavior indicator network during CSGO period.

(Utility—Utility), ODPR—OA (Opening—Opening), KPR—RWMK (Firepower—Firepower), KPR—RWK (Firepower—Firepower), SKP—RWSKP (Sniping—Sniping). Top 11–20 strongly associated edges followed by ODPR—OS (Opening—Opening), DPRW—DPK (Firepower—Trading), OKPR—OS (Opening—Opening), KPRW—SR (Firepower—Entrying), SR—SPRL (Entrying—Clutching), FTPR—TOFPR (Utility -Utility), ASPR—ATPR (Entrying—Opening), CPPR—LAP (Clutching—Clutching), SBTPR—STPR (Entrying—Trading), KPRW—DPRW (Firepower—Firepower). The network of in-game behavior during the CS2 period is shown in Figure 2. Top ten strongly associated edges of the in-game behavior network during the CS2 period followed by TKPR—TKP (Trading—Trading), FTPR—FAPR (Utility—Utility), SKP—RWSKP (Sniping—Sniping), TDPR—TDP (Entrying—Entrying), OKPR—OA (Opening—Opening), KPRW—DPRW (Firepower—Firepower), ODPR—OA (Opening—Opening), UDPR—UKPR100 (Utility—Utility), ODPR—OS (Opening—Opening), OKPR—OS (Opening—Opening). Top 11–20 strongly associated edges followed by KPR—RWMK (Firepower—Firepower), TAPR—SPRL (Clutching—Clutching), KPR—RWK (Firepower—Firepower), FAPR—TOFPR (Utility—Utility), ASPR—DPK (Entrying -Trading), KPR—DPR (Firepower—Firepower), CPPR—LAP (Clutching -Clutching), SR—SPRL (Entrying—Clutching), KPR—DPK (Firepower—Trading), SKP—SMKR (Sniping—Sniping).

In Table 1 and Figure 2, during the CSGO period, SR (Entrying) exhibited the strongest node strength, followed by ASPR (Entrying), LAP (Clutching), DPR (Firepower), DPK (Trading). RWSKP (Sniping) exhibited the strongest Predictability, followed by SKP (Sniping), KPR (Firepower), OA (Opening), SMKR (Sniping). TDP (Entrying) exhibited the strongest expected influence, followed by TKPR (Trading), KPR (Firepower), FAPR (Utility), OKPR (Opening). In Figure 3, DPR (Firepower) exhibited the strongest bridge strength, followed by LAP (Clutching), AKP (Trading), WAOK (Opening), KPRW (Firepower). In Table 2 and Figure 4 during the CS2 period, SR (Entrying) exhibited the strongest node strength, followed by DPK (Trading), ASPR (Entrying), TDP (Entrying), DPR (Firepower). KPR (Firepower) exhibited the strongest Predictability, followed by OA (Opening), SKP (Sniping), DPRW (Firepower), ODPR (Opening). TDP (Entrying) exhibited the strongest expected influence, followed by DPR (Firepower), ASPR (Entrying), FAPR (Utility), TKPR (Trading). In Figure 5, SR (Entrying) exhibited the strongest bridge strength, followed by ASPR (Entrying), PRR (Firepower), AKP (Trading), UDPR (Utility).

3.2. Network stability

As shown in Figure 6, we conducted a network stability analysis using the case-dropping bootstrap implemented in the bootnet package. The results indicated that the correlation-stability coefficients

Table 1. Network metrics of the in-game behavior during CSGO period.

In-game behavior indicators	Strength	Expected influence	Predictability	Community
Firepower				
KPR	2.120	1.492	0.980	1
KPRW	1.880	0.128	0.910	2
DPR	2.180	1.241	0.911	2
DPRW	1.899	1.168	0.878	2
RWK	1.630	0.620	0.914	1
RWMK	1.690	0.886	0.933	1
PRR	1.391	0.159	0.385	2
Entrying				
SBTPR	1.749	0.386	0.766	2
TDPR	1.894	0.649	0.957	2
TDP	2.055	1.667	0.926	2
ODTP	0.971	0.730	0.797	2
ASPR	2.479	0.782	0.875	2
SR	2.641	0.620	0.838	2
Trading				
STPR	1.739	0.572	0.763	2
TKPR	1.668	1.522	0.956	1
TKP	1.649	0.878	0.954	1
AKP	2.124	0.115	0.746	2
DPK	2.176	0.486	0.846	2
Opening				
OKPR	1.543	1.311	0.961	3
ODPR	1.691	−0.124	0.951	3
OA	1.626	1.167	0.977	3
OS	1.762	0.433	0.896	3
WAOK	1.440	−0.030	0.360	2
ATPR	1.589	0.943	0.868	2
Clutching				
CPPR	1.239	0.439	0.484	2
LAP	2.262	0.251	0.779	2
WP1V1	0.875	0.183	0.110	2
TAPR	1.148	0.663	0.623	2
SPRL	1.964	0.244	0.724	2
Sniping				
SKPR	1.024	0.852	0.966	4
SKP	1.289	1.130	0.982	4
RWSKP	1.259	1.229	0.988	4
SMKR	0.905	0.875	0.968	4
SOKPR	0.953	0.577	0.936	4
Utility				
UDPR	1.656	1.248	0.765	5
UKPR100	1.208	0.911	0.665	5
FTPR	1.771	0.585	0.685	5
FAPR	1.779	1.444	0.854	5
TOFPR	1.624	0.458	0.717	5

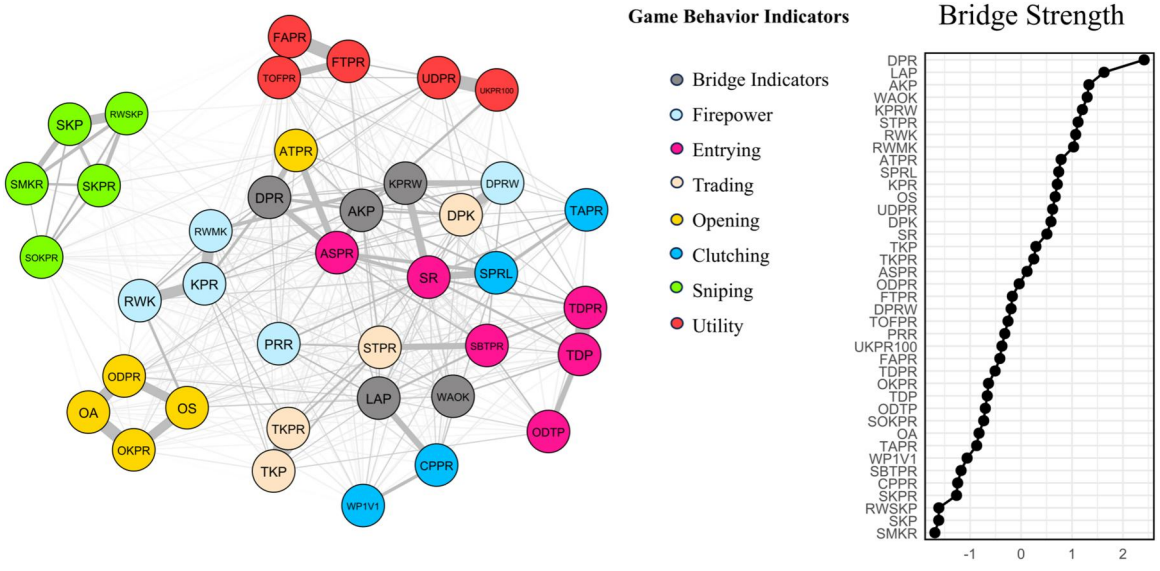


Figure 3. Bridge nodes of in-game behavior indicator network during CSGO period.

Table 2. Network metrics of the in-game behavior during CS2 period.

In-game behavior indicators	Strength	Expected influence	Predictability	Community
Firepower				
KPR	2.149	1.122	0.986	1
KPRW	1.734	0.565	0.958	1
DPR	2.158	1.556	0.938	1
DPRW	1.786	0.900	0.961	1
RWK	1.462	0.506	0.942	1
RWMK	1.377	0.800	0.948	1
PRR	1.305	0.221	0.461	2
Entrying				
SBTPR	1.291	0.845	0.778	2
TDPR	1.992	0.154	0.927	2
TDP	2.210	1.656	0.918	2
ODTP	0.910	0.564	0.656	2
ASPR	2.243	1.432	0.884	1
SR	2.572	0.581	0.852	2
Trading				
STPR	1.236	0.884	0.797	3
TKPR	1.450	1.370	0.955	3
TKP	1.765	0.726	0.953	3
AKP	1.863	0.144	0.753	1
DPK	2.405	0.084	0.947	1
Opening				
OKPR	1.725	1.297	0.953	4
ODPR	1.691	−0.195	0.961	4
OA	1.570	1.162	0.979	4
OS	1.751	0.308	0.917	4
WAOK	1.071	0.093	0.196	2
ATPR	1.507	0.859	0.862	1
Clutching				
CPPR	1.072	0.746	0.527	2
LAP	1.591	0.031	0.783	2
WP1V1	1.060	0.260	0.125	2
TAPR	1.478	0.686	0.711	2
SPRL	1.917	0.586	0.672	2
Sniping				
SKPR	1.197	0.891	0.899	5
SKP	1.407	1.169	0.961	5
RWSKP	1.128	1.103	0.957	5
SMKR	0.944	0.771	0.879	5
SOKPR	0.927	0.545	0.797	5
Utility				
UDPR	1.665	1.243	0.733	6
UKPR100	1.179	0.783	0.495	6
FTPR	1.475	0.641	0.688	6
FAPR	2.030	1.375	0.821	6
TOFPR	1.216	0.360	0.308	6

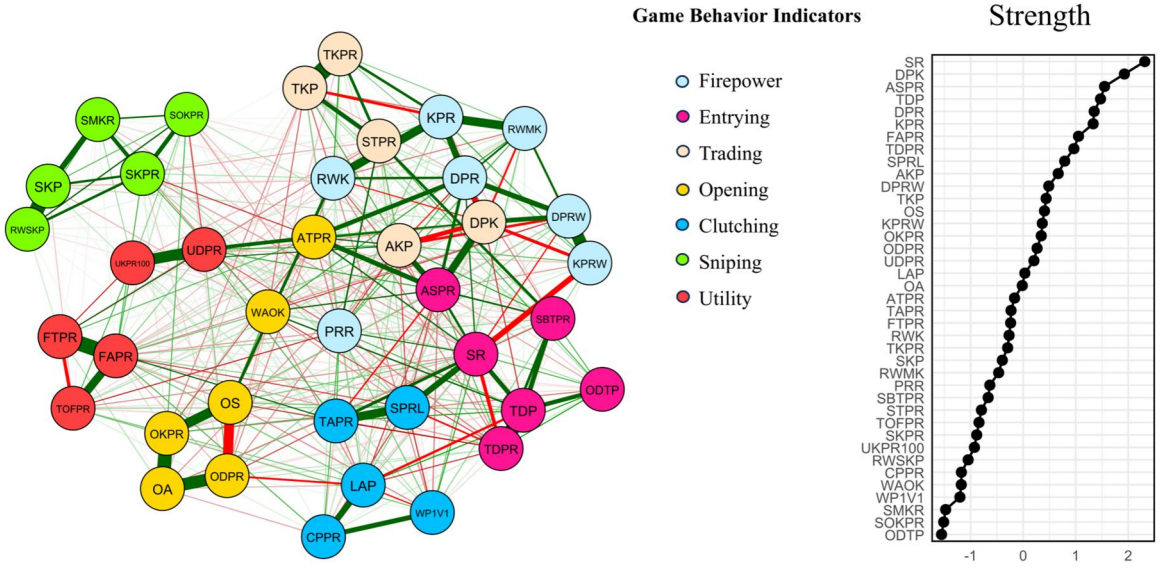


Figure 4. In-game behavior indicator network during CS2 period.

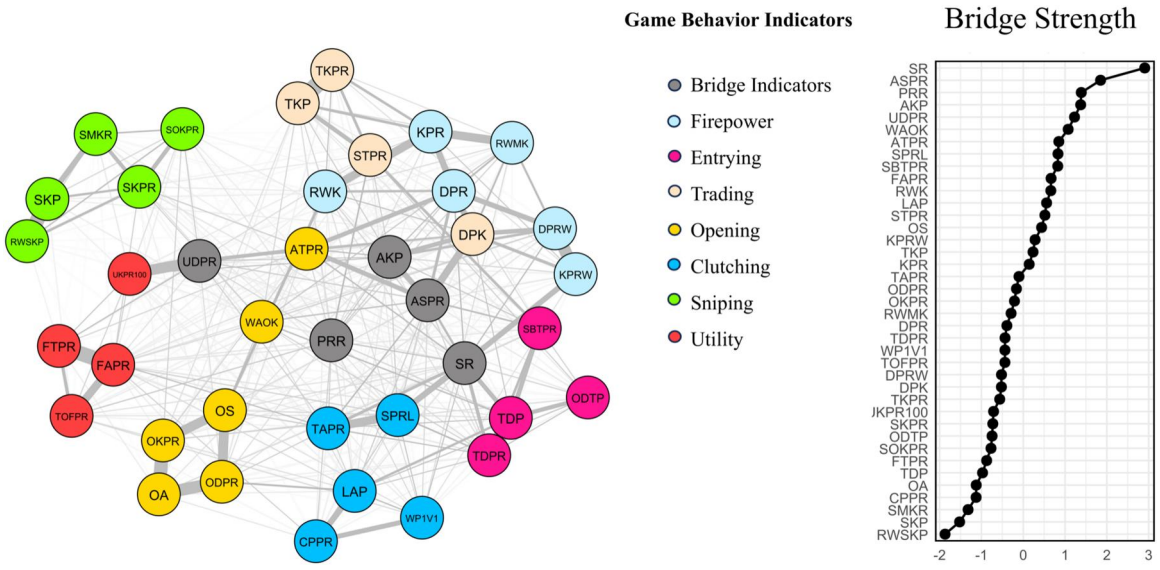


Figure 5. Bridge nodes of in-game behavior indicator network during CS2 period.

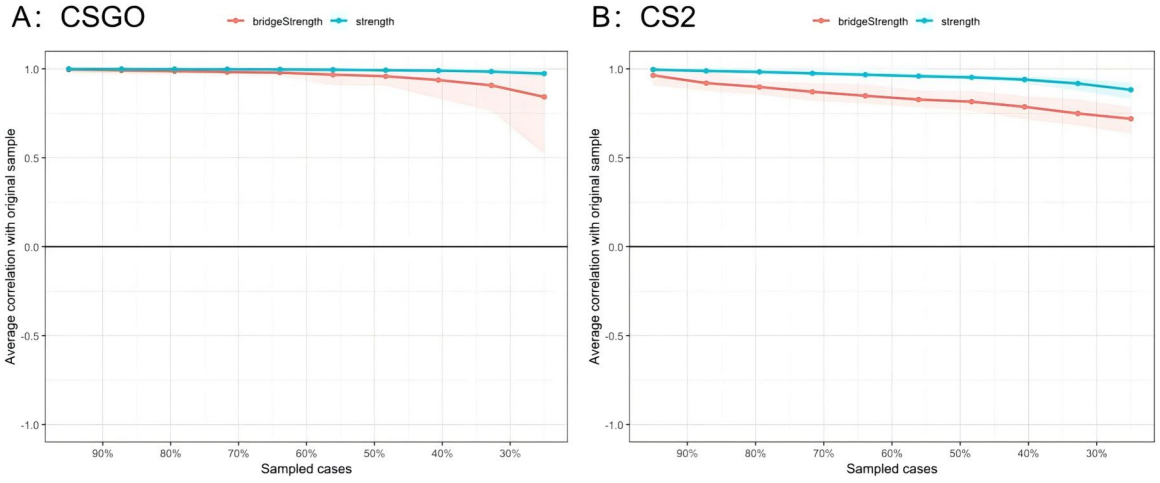


Figure 6. The in-game behavior indicator network stability test of strength and bridge strength by using case-dropping bootstrap during CSGO and CS2 periods. The horizontal axis represents the proportion of the original sample retained, and the vertical axis represents the average correlation coefficient between the initial network index and the recalculated centrality index after case culling.

(CS-C) for the centrality network metrics, node strength, and bridge strength were 0.75 and 0.67 during the CSGO period. Node strength and bridge strength were 0.75 and 0.59 during the CS2 period. It all exceeded the recommended value of 0.5, suggesting that these metrics possess adequate stability. Pairwise differences in node strength detail were presented in Figure 7. Additionally, the bootstrapped 95% confidence intervals (CIs) for the edge weights' result shown in supplemental material for reference.

3.3. Network comparison

We conducted a comparative analysis of network invariance across the CS:GO and CS2 periods, focusing on global network structure invariance and global network strength. Edges invariance and centrality invariance will be provided in supplemental materials for reference.

As shown in Figure 8, there was significant difference in network global strength (CSGO: 31.537 vs CS2: 29.577, $S = 1.960$, $p < 0.001$) during two periods. There was significant difference in network structure invariance (Test statistic $M = 0.483$, $p < 0.001$) during two periods. Due to space limitations, details

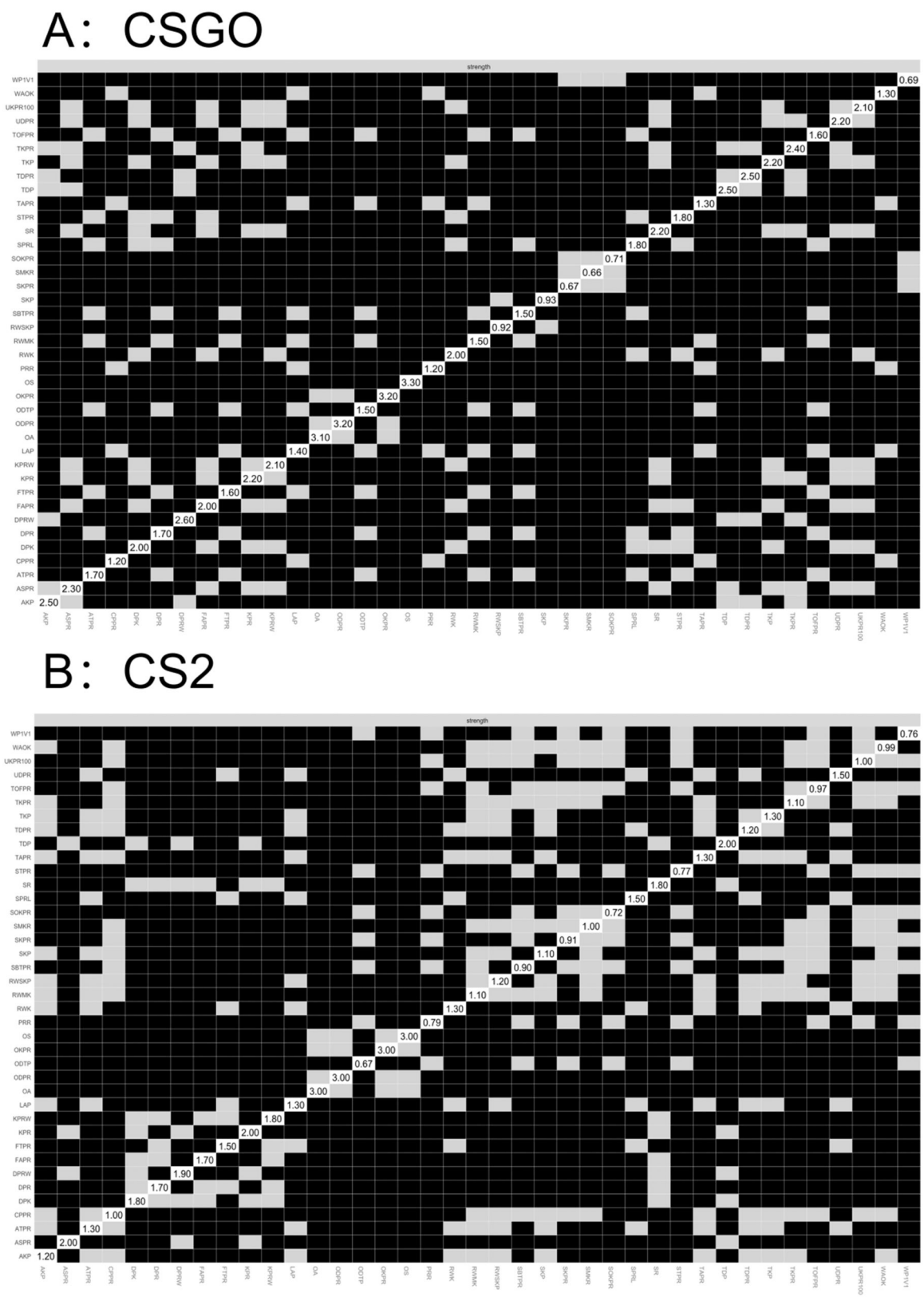


Figure 7. Pairwise differences in node strength among in-game behavior indicator network during CSGO and CS2 periods. Gray boxes indicate no significant differences, and black boxes indicate significant differences. Numbers in white boxes (i.e., diagonal lines) indicate node strength values for specific nodes.

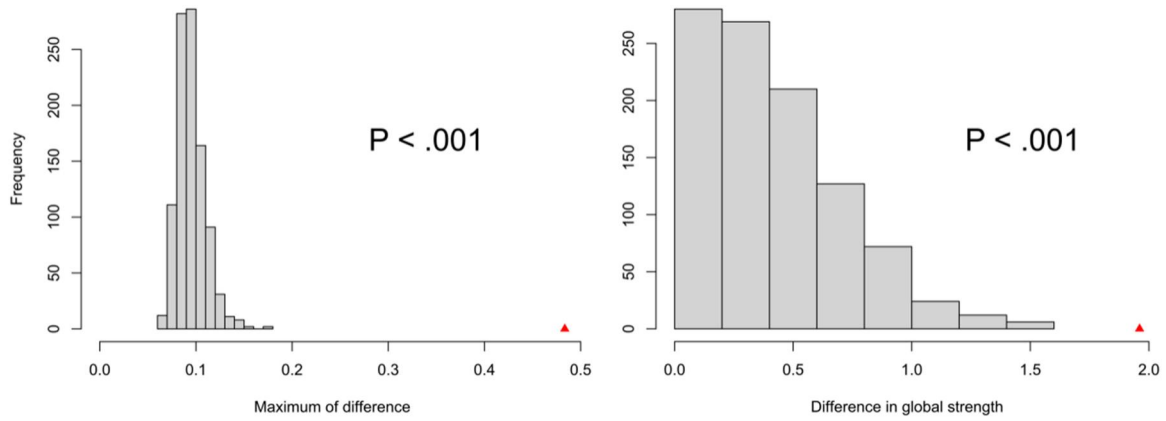


Figure 8. Network comparison test of global strength and network structure among CSGO and CS2 periods. The red triangle indicates the observed test statistic derived from the original data. The gray histogram represents the null distribution of the test statistic, generated through repeated permutation of group labels. If the red triangle falls into the tail of the distribution and the *p*-value is below 0.05, the between-group difference is considered statistically significant.

of the edge invariance test will be provided in the supplementary materials. The results of the centrality invariance test as shown in the supplementary materials, also indicate that the relative importance of certain key nodes in the in-game behavior networks across the two periods has changed.

3.4. Network community

Community detection based on the Spinglass algorithm exhibits distinct segmentation from official in-game behavior indicator dimensions across both CSGO and CS2 network phases as shown in Figure 9.

During the CS:GO period, five distinct communities of in-game behavior indicators emerged. The first community comprised KPR (Firepower), RWK (Firepower), RWMK (Firepower), TKPR (Trading),TKP (Trading). The second community comprised KPRW (Firepower), DPR (Firepower), DPRW (Firepower), PRR (Firepower), SBTPR (Entrying), TDPR (Entrying), TDP (Entrying), ODTP (Entrying), ASPR (Entrying), SR (Entrying), STPR (Trading), AKP (Trading), DPK (Trading), WAOK (Opening), ATPR (Opening), CPPR (Opening), LAP (Clutching), WP1V1 (Clutching), TAPR (Clutching), SPRL (Clutching). The third community comprised OKPR (Opening), ODPR (Opening), OA (Opening), OS(Opening). The fourth community comprised SKPR (Sniping), SKP (Sniping), RWSKP (Sniping), SMKR (Sniping), SOKPR (Sniping). The fifth community comprised UDPR (Utility), UKPR100 (Utility), FTPR (Utility), FAPR (Utility), TOFPR (Utility).

During the CS2 period, six distinct communities of in-game behavior indicators emerged. The first community comprised KPR (Firepower), KPRW (Firepower), DPR (Firepower), DPRW (Firepower), RWK (Firepower), RWMK (Firepower), ASPR (Entrying), AKP (Trading), DPK (Trading), ATPR (Opening). The second community comprised PRR (Firepower), SBTPR (Entrying), TDPR (Entrying), TDP (Entrying), ODTP (Entrying), SR (Entrying), WAOK (Opening), LAP (Clutching), WP1V1 (Clutching), TAPR (Clutching), SPRL (Clutching), LAP (Clutching). The third community comprised TKPR (Trading), TKP (Trading), STPR (Trading). The fourth community comprised OKPR (Opening), ODPR (Opening), OA (Opening), OS(Opening). The fifth community comprised SKPR (Sniping), SKP (Sniping), RWSKP (Sniping), SMKR (Sniping), SOKPR (Sniping). The sixth community comprised UDPR (Utility), UKPR100 (Utility), FTPR (Utility), FAPR (Utility), TOFPR (Utility).

It is important to note that the community segmentation in both periods exhibited distinct communities compared to the originally official in-game behavior indicator dimensions. Only the segmentation and composition of in-game behavior indicators in the sniping and utility dimensions remained unchanged across the two periods. Furthermore, each community in the detection results from both periods contained at least one strongly connected edge, and in-game behavior indicator nodes with high bridging strength occupied a core hub position in connecting different communities.

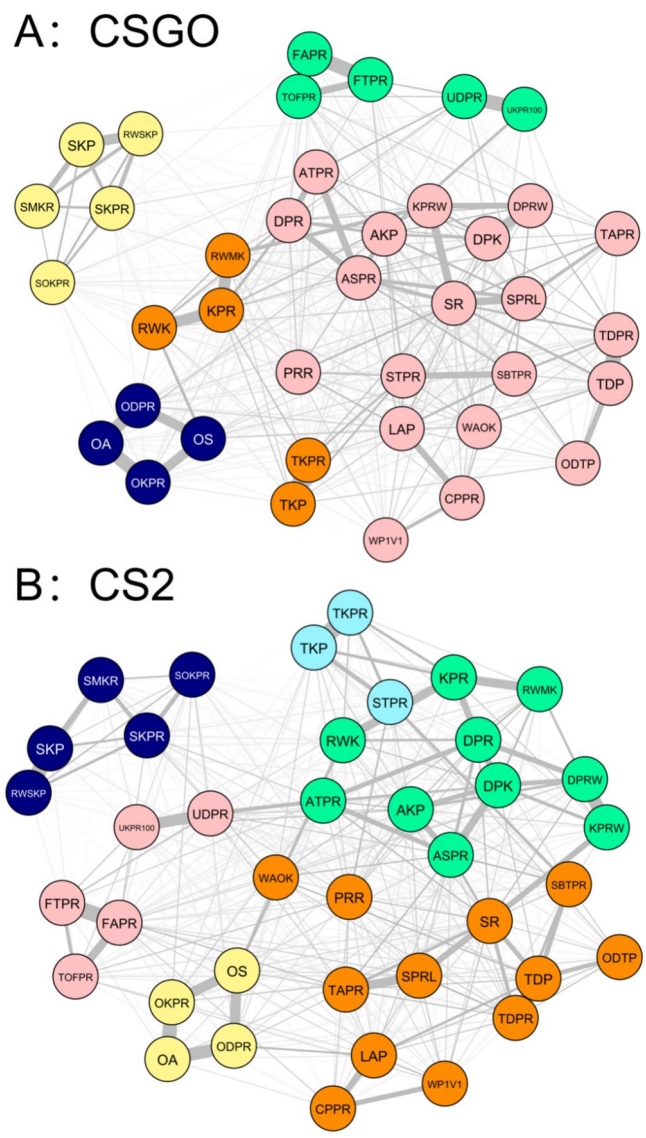


Figure 9. Community detection among in-game behavior indicator network during CSGO and CS2 periods. The colors represent different community clusters. Edge colors are removed to avoid confusion.

4. Discussion

In the in-game behavior networks of CS:GO and CS2, the top ten strong edges primarily cluster within the same in-game behavior dimension. This indicates that indicators exhibit clear intra-dimensional clustering at the network level. However, strong cross-dimensional associations such as SR–SPRL (Entry–Clutching) begin to emerge among the 11th–20th ranked edges. This suggests cross-structural connections between different in-game behaviors across distinct tactical capabilities. This may indicate a stable conditional association between rounds characterized by assist capabilities and rounds characterized by resource conservation (Xenopoulos et al., 2021). This also reflects the behavioral co-occurrence of economic management strategies in professional matches (Sharpe et al., 2023; Xenopoulos et al., 2021). Simultaneously, multi-dimensional associations between firepower and trading indicators suggest structural connections between output mechanisms, kill efficiency, and follow-up shot scenarios (HLTV, 2023). Professional CS players' performance is not solely determined by kill counts, in-game behaviors like damage patterns and kill efficiency may be more deeply embedded within systems of team coordination and resource expenditure (HLTV, 2023; Sharpe et al., 2023).

Network node findings across the two periods exhibit both similarity and differences. First, SR (Entrying) demonstrated the highest node strength in both the CS:GO and CS2 periods, while TDP

(Entrying) showed the highest expected influence in both stages. Within the network structure, high node strength aligns closely with the structural implication of extensive direct conditional connections with other nodes in the system (Bringmann et al., 2019). Consequently, SR (Entrying)'s persistent central position may relate to the comprehensive nature of its operational definition. SR (Entrying) inherently involves multiple rounds of events, including assists, survival, and being traded out, which establishes direct links with several tactical modules (Jing et al., 2025). TDP's high expected influence suggests that the "swap-out structure" holds cross-module pivotal significance within the professional performance system, potentially spanning multiple gameplay tactical phases, including initial engagements, damage trading, and sacrificial entry (HLTV, 2024). Additionally, the consistently high node strength of ASPR, DPK, and DPR across two periods indicates that in-game behaviors related to output and damage are also highly important. This is consistent with previous research findings, which indicate that effective offensive actions and sustained damage output remain the core behavioral characteristics of first-person shooter (FPS) games (Jing et al., 2025; Nagorsky & Wiemeyer, 2020).

Predictability results further reveal that the embedding degree of different in-game behavior indicators within network structures may change with version updates and shifts in tactical and gameplay trends (Thaicharoen et al., 2023). During the CS:GO period, sniper-related indicators (e.g., RWSKP, SKP, SMKR) exhibited high predictability. In contrast, the CS2 period showed higher predictability for indicators like KPR and OA, which reflect more holistic output and opening engagement. High predictability typically indicates that the indicator can be largely explained by other neighborhood indicators within the network (Haslbeck & Waldorp, 2018). This suggests that sniper behavior in CS:GO may be more tightly embedded within other tactical performance structures, whereas CS2 exhibits stronger systemic couplings between output behavior and opening engagement. This distinction helps clarify how in-game behavior are organized and constructed during version updates and tactical evolution.

The discovery of bridge strength provides more direct evidence for the restructuring of the associative architecture of in-game behavior indicators. During the CS:GO period, DPR (Firepower) exhibited the highest bridge strength, suggesting that "damage per round" plays a pivotal hub between different tactical modules. In the CS2 period, SR (Entry) emerged as the strongest bridge node, while UDPR (Utility) advanced into the top tier of bridge nodes. This shift indicates that the "structural bonds" connecting in-game behavior modules may be partially transitioning from output-oriented damage behaviors toward support-oriented structures and in-game utility usage. Previous research has confirmed the critical role of teamwork and utility usage in modern esports performance (Jing et al., 2025). During the CS2 period, the prominence of non-firepower and non-damage related in-game behaviors underscored the importance of tactical diversity and evolving playstyles in esports (Gisbert-Pérez et al., 2024; Jing et al., 2025; Nagorsky & Wiemeyer, 2020). The timing and deployment of smoke grenades and flashbangs are closely linked to professional players' tactical decisions and team coordination (Horst et al., 2021; Llvarinen, 2022; Oksala, 2021). Furthermore, the bridge role played by AKP (Trading) in both periods suggests that coordinated kills with teammates and collaborative damage output may be relatively stable components of cross-module connectivity. This consistently reflects the key structural importance of teamwork-based in-game behaviors and performance within the professional CS competitive ecosystem (HLTV, 2023; Jing et al., 2025).

Network centrality metrics in network analysis must exhibit sufficient robustness (Fried et al., 2022). Otherwise, their interpretations may reflect results influenced by sample fluctuations rather than actual structural characteristics (Epskamp et al., 2018). Results indicate that during both CS:GO and CS2 periods, the correlation stability coefficients for node strength and bridging strength both exceeded the recommended threshold of 0.50 (Epskamp et al., 2018). This finding indicates that the core centrality node metrics in this study exhibit good structural robustness and interpretive stability. The estimation uncertainty of edges is typically higher than that of the overall structure or centrality patterns, especially when edge weights are small or node relationships are weak. Therefore, reporting 95% confidence intervals for edge weights helps avoid overinterpreting individual weak edges when analyzing cross-version differences (Borsboom et al., 2021).

The results of the network comparison test indicate that the in-game behavior indicator networks of CS:GO and CS2 exhibit certain differences. Specifically, the two periods show significant differences in

global network strength, and the network structure invariance test also reveals significant differences. From a network perspective, global strength reflects the overall level of all edge weights in the network (Epskamp et al., 2018). Therefore, this result suggests that the overall connectivity of the networks has changed between the two periods. The differences in network structure invariance further indicate that not only has the strength changed, but the conditional association patterns between in-game behavior indicators themselves have also been reorganized. It is worth noting that the overall network strength during the CS2 period was significantly lower than that during the CS:GO period. This may reflect a greater differentiation in the relationships and gameplay style between in-game behavior indicators within new game ecosystem (Thaicharoen et al., 2023). In general, the evolution of game versions, balance adjustments, and tactical trends has been accompanied by structural reorganization at the systemic level of in-game behavior indicators (Thaicharoen et al., 2023). This reorganization cannot be fully captured by a single indicator alone.

Community detection results indicate that the segmentation of gaming behavior indicator communities based on the Spinglass algorithm exhibits differences from HLTV's official dimensions across both CS:GO and CS2 periods. This finding implies that while official classifications rely on operational definitions to categorize seven indicator dimensions, "Firepower, Entrying, Trading, Opening, Clutching, Sniping, and Utility" (HLTV, 2024), actual in-game behavior indicator in competitive scenarios more likely manifests as cross-dimensional behavioral modules. This suggests that as the game ecosystem evolves, the correlations between in-game behavioral indicators across different games will fluctuate (Zhang & Keegan, 2022). However, the specific reasons should be comprehensively considered based on version updates, tactical developments, and current gaming trends. Therefore, community segmentation based on in-game behavior during version updates may not fully align with previous dimensions. The reorganization of indicator's dimensions primarily occurs within the Firepower, Entrying, Trading, Opening, and Clutching. Compared to the CS:GO period, the CS2 period exhibits more detailed modular segmentation in community detection. This suggests that, influenced by changes in game mechanics or balance resulting from the evolution of the gaming environment, in-game behavioral dimensions may develop more differentiated organizational structures as new strategies to adapt (Thaicharoen et al., 2023). Certain in-game behavior chains (e.g., fire/attack events linked to kill efficiency and coordinated kills) form more concentrated modules within the network. During these two periods, the community segmentation based on the sniping dimension and utility dimension consistently aligned with original dimensions. This may indicate that a structural association exists between in-game behaviors related to sniping ability and utility usage. It may also reflect the role specialization and recognizability of these behaviors in professional esports competitions (Shevchuk, 2024; Young, 2021). Therefore, community detection reveals a structure where certain dimensions remain independent, while some cross-dimensional indicators reorganize to form large-scale in-game behavior communities.

From the perspective of the digital competitive gaming environment, we believe that the findings of our network analysis offer the following practical implications for training priorities, tactical adjustments, and player selection. First, Support Rounds (Entrying) and Traded Deaths Percentage (Entrying) possess the highest node strength and expected influence during the two periods. Therefore, esports coaches and data analysts should refocus and monitor their priorities on these highly network-centralized and team-beneficial behaviors, rather than selecting players based solely on their individual damage performance. This may indicate that the supportive behaviors of adhesive or team-oriented players play a structurally critical role in the overall in-game behavioral network. High-strength in-game behavior should also be considered the most worthy training part for refining and prioritization. The dense intra-dimensional connections observed among Entrying, Trading, and Firepower indicate that these behaviors should not be trained in isolation. Daily practice and tactical coordination should focus on combining cross-action patterns. For example, experimenting with different timing for deploying support utilities and then immediately following up with a coordinated trading or breakthrough to see the actual benefits. Second, when evaluating roles and configuring team compositions, greater emphasis should be placed on interaction between support coordination and utility-linked behaviors. The bridge indicators should be treated as supplementary metrics for assessing a player's "cross-module integration capabilities." When a player's core performance metrics fluctuate or decline, predictability can serve as

an assessment and diagnostic tool based on its network characteristics. By analyzing the level of predictability, it is possible to determine the causes of fluctuations influenced by other in-game behaviors or performance factors, as well as identify potential improvement pathways. In addition, this network perspective can help game developers update and reshape the ecosystem of gameplay interactions, enabling them to timely identify issues in the current game version. It should be noted that these insights require practical adjustments and testing in a sufficient number of real esports matches to obtain authentic feedback.

This study provides systematic evidence on the structural organization and topological characteristics of in-game behavior networks within esports environments. Compared to traditional statistical approaches, the network perspective reveals several issues concerning professional esports performance indicators in terms of network structure, modular communities, and cross-module connectivity mechanisms. In the dynamic landscape of esports, where game versions, balance adjustments, technical platforms, and tactical trends are constantly evolving (Thaicharoen et al., 2023), the reconfiguration of players' in-game behavioral dimensions offers a fresh perspective for understanding esports in-game behavior and performance. However, this study has several limitations. First, edges in the network represent conditional associations in a partial correlation context rather than causal relationships between in-game behaviors. Second, although the comparison of the two game phases reveals significant differences, CS2 is currently still in development and undergoing data updates. Therefore, the data must be interpreted within the context of the current timeline.

For future research, it is advisable to continue tracking and incorporating more longitudinal data with updates. By combining with qualitative interviews, recording data on daily practice sessions may also offer a different analytical perspective (Bennett et al., 2026; Boldi et al., 2025). This would better facilitate revealing the structural evolution trajectories of player behavior during game development and individual improvement processes. There may also be some interesting insights to be gained from analyzing the specific behavioral patterns of overlapping player groups across different periods. In addition, future research could incorporate contextual information such as team roles (sniper, rifleman, commander), map features, and competition level. This will help expand network-based esports research toward a broader understanding of behavioral evolution within digital competitive environments.

5. Conclusion

In summary, this study demonstrates that the in-game behavior network of professional Counter-Strike players constitutes a conditionally interconnected system integrated from multiple behavioral modules. The evolution of the game has not only brought about profound changes in the overall connectivity and structural patterns of in-game behaviors, but also redefined the structure of behavioral communities and evolved cross-module connection mechanisms. By applying network analysis, this study provides a novel network perspective for understanding the interconnected structures and systematic organization of professional esports players' in-game behaviors. This also provides an empirical foundation for future research exploring the dynamics of version iterations, role specialization, tactical trends, as well as the evolution of the in-game behavior ecosystem within the esports environment.

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Author contributions

CRediT: **Shuoyu Jing**: Conceptualization, Formal analysis, Methodology, Validation, Writing – original draft, Writing – review & editing; **Mohd Mahzan Awang**: Conceptualization, Supervision, Writing – review & editing; **Wan Ahmad Munsif Wan Pa**: Conceptualization, Supervision, Writing – review & editing; **Pengyu Pan**: Conceptualization, Formal analysis, Visualization, Writing – original draft, Writing – review & editing; **Zhibin Zhang**: Formal analysis, Methodology, Visualization, Writing – original draft, Writing – review & editing.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

Process files can be obtained from the supplementary materials. The raw data can be obtained from GitHub link via supplementary materials.

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